ZAIN UL ABIDIN

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ABOUT ME

Highly focused and enthusiastic Computer Systems Engineering student with a strong passion for Artificial Intelligence and Machine Learning, particularly in computer vision and image processing. Experienced in developing ML/DL frameworks for defect detection and image super-resolution. Committed to advancing innovative AI solutions through graduate research.

EDUCATION

Bachelor of Engineering in Computer Systems Engineering
Sukkur IBA University [09/2022 — 06/2026]
Address: Sukkur, Sindh, Pakistan | https://www.iba-suk.edu.pk
Final Grade: 3.77 out of 4.00
Accreditation: PEC Accredited Program – Level-II (Washington Accord Substantial Equivalence)
Relevant Courses (Grades): Artificial Intelligence & Machine Learning (A), Digital Signal Processing (A), Signals & Systems (A), Data Structures & Algorithms (A), Object-Oriented Programming (A-), Computer Programming (A-), Discrete Structures (A-), Probability and Statistics (A), Linear Algebra & Differential Equations (A), Database Management Systems (A-), Digital Logic Design (A)

WORK EXPERIENCE

Artificial Intelligence Intern
Mirai School of Technology | Jul 2025 – Aug 2025 (Remote)

- Built Conversational Order Assistant using Agentic AI workflows (n8n, Gemini AI).
- Automated order capture, validation, logging, and email confirmations via APIs.
- Gained practical experience in ML, prompt engineering, and AI automation.

Software Engineering Intern
Sukkur Electric Power Company (SEPCO) | Jun 2025 – Jul 2025 (On-Site, Sukkur, Sindh)

- Developed Mute Meter Monitoring Dashboard (Python, Streamlit, SQLite) with role-based access.
- Implemented GIS filters and bulk import/export, reducing manual inspection by 40%.
- Digitized meter monitoring for enhanced efficiency.

PUBLICATIONS

Published

- Zain-ul-Abidin, Haseeb. *Mitigation of DDoS Attacks with Machine Learning, Deep Learning and Transformers.* International Journal of Multidisciplinary Conference Proceedings (IJMCP), Vol. 2, No. 1, ICISCT'25, July 2025. https://doi.org/10.61503/Ijmcp.v2i1.181

Under Review

- Zain Ul Abidin, Junaid Ahmed, Dong Wang, Guiyun Tian. *Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model.* **Nondestructive Testing and Evaluation (Taylor & Francis)**, Under Review (Submission ID: 255311532).
- Faisal Sahito, Aakash Kumar, Dr. Junaid Ahmed, Zain Ul Abidin, Dr. Pan Zhiwen. *Improving Image Super-Resolution: A Hybrid Approach Using DT-CWT and Transformer Networks.* **Image and Vision Computing (Elsevier)**, Under Review (Manuscript ID: IMAVIS-D-25-03665).

HONORS & ACHIEVEMENTS

- Prime Minister's Laptop Award (Merit-Based)
- Sindh Educational Endowment FUND (SEEF) Scholarship – Merit-Based Tuition Coverage
- Consistently ranked among top students.

ACADEMIC & RESEARCH PROJECTS

Final Year Project – Subsurface Defect Detection in CFRP Using Optical Pulse Thermography (In Progress)

- Developing ML/DL framework for defect detection in CFRP using pulsed thermography datasets.
- Exploring architectures (Cascade R-CNN, DeepLabv3+, Transformers) for cross-dataset validation.

Gold Price Prediction System

- Built forecasting dashboard with XGBoost, LSTM, and Random Forest.
- Deployed using Streamlit with trend graphs, CSV input, and export features.

Speech Emotion Recognition (LSTM)

- Designed real-time SER system using MFCC, ZCR, RMS, Chroma features with augmentation.
- Achieved **97% accuracy** on TESS dataset with robust LSTM model.

Real-Time Speech Enhancement

- Implemented denoising via Spectral Subtraction & Wiener Filtering in Python.
- Improved speech clarity with up to 10 dB SNR gain while preserving intelligibility.

Hand Gesture Recognition & Voice Conversion

- CNN-based system translating hand gestures to speech for assistive communication.
- Achieved ~90% recognition accuracy for deaf/mute support.

Student Management System with ML Integration

- Tkinter + MySQL app with role-based portals for admins, teachers, students, and parents.
- Integrated predictive analytics with ML models for performance tracking.

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Google IT Support Professional Certificate, Google Career Certificates, Coursera [2025]

- Gained hands-on experience in computer hardware, troubleshooting, and system maintenance.
- Configured networks (TCP/IP, DNS, DHCP, VPNs) and managed cloud and virtualized systems.
- Practiced Windows/Linux administration with focus on security, recovery, and automation.

Google Prompting Essentials Specialization, Google Career Certificates, Coursera [2025]

- Developed prompt engineering skills using Google's 5-step prompting framework.
- Applied generative AI tools (ChatGPT, Gemini) for summarization, ideation, and data analysis.
- Practiced responsible AI use through bias detection, prompt refinement, and versioning.

Machine Learning Specialization, DeepLearning.AI, Coursera [2025]

- Built supervised and unsupervised ML models using Python, TensorFlow, and PyTorch.
- Applied ensemble methods (random forests, boosting) and neural network architectures.
- Designed recommender systems and reinforcement learning models for real-world tasks.

Mathematics for Machine Learning and Data Science Specialization, DeepLearning.AI, Coursera [2025]

- Strengthened foundations in linear algebra, calculus, probability, and statistics for ML.
- Applied gradient-based optimization and analytical methods to model training.
- Utilized NumPy and Python for data-driven mathematical modeling and PCA-based analysis.

ChatGPT Prompt Engineering for Developers, DeepLearning.AI, Coursera [2025]

- Learned to integrate and optimize large language models (LLMs) using OpenAI APIs.
- Engineered effective prompts for summarization, inference, and text transformation.
- Built and tested custom AI chatbots through Jupyter-based coding exercises.

SKILLS AND ABILITIES

Research & Lab: ML/DL experiments, image & signal processing, speech analysis, feature extraction
Software & Tools: Python, MATLAB, Streamlit, Jupyter, Git/GitHub, MySQL, SQLite
Libraries & Frameworks: NumPy, Pandas, SciPy, Matplotlib, Seaborn, TensorFlow, PyTorch, OpenCV, PyWavelets
AI/ML Applications: Supervised, unsupervised learning, object detection, image super-resolution, transformers

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RESEARCH INTERESTS

Artificial Intelligence · Machine Learning · Image Processing · Audio Processing · Digital Signal Processing · Intelligent Systems · Computer Vision